S43 | Sparse render-and-compare hand tracking

W50

Fuse event evidence and the projected previous hand to estimate a 51-D state correction

(a) Event evidence and prior

Previous state $\mathbf{x}_{k-1} \in \mathbb{R}^{51}$



MANO mesh

778 vertices



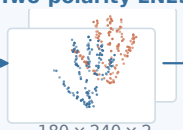
Vertex projection

Occupancy / inv. depth

50-ms events



(x, y, t, p)



Two-polarity LNES

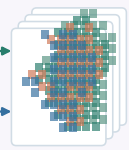
$180 \times 240 \times 2$

(b) Sparse encoder and readout

Previous-state bypass

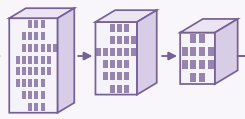
Support = LNES ∪ projection

3×3 adapter 4→3 + coordinates



4-channel fusion

Sparse ResNet18-L3



64 128 256

2 blocks / stage · MaskedBN

Active mean ‖ area-scaled sum

256 + 256 → FC 512→51

(c) Recurrent state update

State correction MLP

51 → 64 → 51

Δ_{CNN} Δ_{MLP} Δ

Residual update + empty hold

$\mathbf{x}_k = \mathbf{x}_{k-1} + g_k \Delta$

$g_k = \mathbf{1}[\|L_k\|_1 > 0]$

State $\mathbf{x}_k \in \mathbb{R}^{51}$

Direct state feedback · one update every 50 ms

51-D = translation 3 + root axis-angle 3 + finger axis-angles 45. Geometry and events are schematic.